MNQOBI BHEBHE
Exercise 3
BRIGHTLEARN TUTORIAL

1. Classify each product by price:
'Expensive' if price>1000
'Mid-range' if price between 100 and 1000
'Budget' if <100

Expected Output Columns:
product_name
price
price_category

```
SELECT product_name,
       price,

CASE
    WHEN price>1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'MID-RANGE'
  ELSE 'Budget'
END AS price_category
FROM products;
```

Product_Name	Price	Price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid range
Keyboard	45.00	Budgetl
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Label each order:
'High Value' for orders ≥ 1000
'Medium Value' for 500–999.99
'Low Value' for orders < 500

Expected output columns:
customer_name
amount
order_value_category

```
SELECT  customer_name,
        amount,

CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
    ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

Customer_name	amount	order_value_category
Alice	150.00	Low value
Bob	500.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low value
Ethan	1200.00	High value

3. Categorize employee position

If in 'IT' and salary > 80000 → 'Senior IT'

If in 'HR' and salary > 55000 → 'Experienced HR'

Otherwise → 'Staff'

Expected output columns

emp_name

department

salary

position_level

```
SELECT emp_name,
       department,
       salary,

CASE
    WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
    ELSE 'Staff'
END AS position_level
FROM employees;
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Assign a letter grade
≥ 90 'A'
80-89: 'B'
70-79: 'C'
60-69: 'D'
<60: 'F'

Expected output Columns
student_name
Score
grade

```
SELECT
    student_name,
    score
  CASE
      WHEN score >= 90 THEN 'A'
      WHEN score BETWEEN 80 AND 89 THEN 'B'
      WHEN score BETWEEN 70 AND 79 THEN 'C'
      WHEN score BETWEEN 60 AND 69 THEN 'D'
  ELSE 'F'
END AS grade
FROM students;
```

Student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Label delivery performance
≤ 30 mins: 'Fast'
31 – 60 mins: 'On Time'
60 mins: 'Late'

Expected output columns
delivery_id
delivery_time_minutes
performance

```
SELECT delivery_id,
       delivery_item_minutes,
CASE
    WHEN delivery_time_minutes <= 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'ON Time'
ELSE 'Late'
END AS performance
FROM deliverables;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. Convert priority to labels
3 -> 'High'
2 -> 'Medium'
1 -> 'Low'

Expected output columns
issue_type
priority
priority_label

```
SELECT
    issue_type,
    priority,
  CASE priority
    WHEN 3 THEN 'High'
    WHEN 2 THEN 'Medium'
    WHEN 1 THEN 'Low'
    END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	low
Server down	3	high
Slow system	2	medium
Email error	2	medium
Password reset	1	low

7. Calculate attendance % and classify
≥90% → 'Excellent'
75-89% → 'Good'
<75 → 'Needs Improvement'

Expected output columns
student_id
attendance_percentage
attendance_status

```
SELECT
 student_id
ROUND((days_present * 100,0) / total_days, 2) AS attendance_percentage,
CASE
WHEN (days_present * 100,0) / total_days >= 90 THEN 'Excellent'
WHEN (days_present * 100,0) / total_days BETWEEN 75 AND 89,99
THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90,00%	Excellent
2	60,00%	Needs Improvement
3	96,00%	Excellent
4	50,00%	Needs Improvement
5	100,00%	Excellent

8.

8. Label stock status
0 → 'Out of Stock'
1-5 → 'Low stock'
5 → 'In Stock'

Expected Output columns
product_id
stock_qty
stock_status

```
SELECT  product_id
        stock_qty
  CASE
    WHEN stock_qty = 0 THEN 'out of Stock'
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
  ELSE AS stock_status
FROM products_inventory;
```

Product_id	stock_qty	stock_status
1	5	low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	low stock

9. Classify by size
≥ 25 → 'Large'
10-24 → 'Medium'
<10 → 'Small'

Expected output columns
subject
enrolled_students
class_size_category

```
SELECT subject,
       enrolled_students,
CASE
   WHEN enrolled_students >= 25 THEN 'Large'
   WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
ELSE 'small'
END AS class_size_category
FROM classes
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Apply discount flag:
If payment_method = 'cash' and amount ≥ 200 → 'Eligible for Discount'
otherwise → 'Not Eligible'

Expected output columns
payment_id
payment_method
amount
discount_eligibility.

```
SELECT payment_id,
       payment_method,
       amount,
  CASE
     WHEN payment_method = 'cash' AND amount >= 200 THEN 'Eligible for Discount'
     ELSE 'Not Eligible'
  END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility.
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount